Sensing Test Protocol

1. Before Arriving on Site

1.1 Make sure that the sensors and smartphone controller are sufficiently charged (i.e., a charge of above 80% on each sensor and the phone).

1.2 Make sure that the sensor layout is well understood before conducting the test. If not, please communicate with Sensequake.

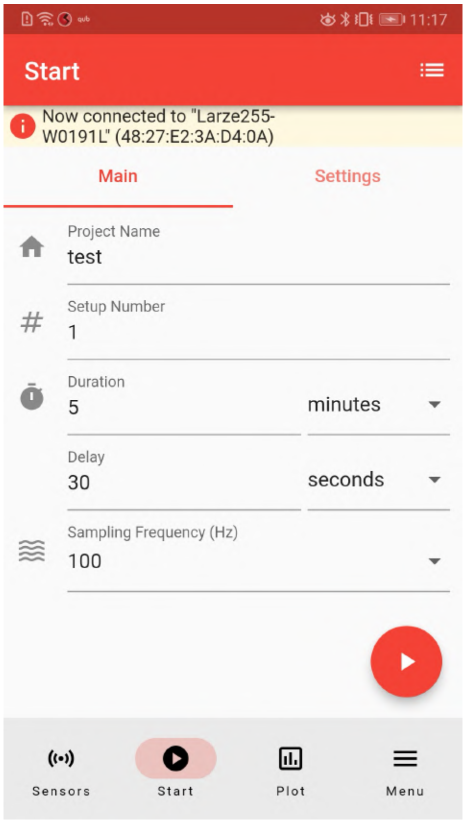
1.3 Perform a sample data measurement a few days before the site test to check that the system is performing as intended.

1.4 Bring the required materials to the site, including but not limited to chalk, measuring tape, measuring wheel, bubble level and sensor antennas.

1.5 If the sensors have been recently shipped, dropped or have not been used in the past 2 months, please conduct the following measurement test and send the data to Sensequake:

1.5.1 Place the sensors in a line as shown in the below image, with the x-direction facing in the same direction. Please take a photo of your setup to be checked by Sensequake.

1.5.2 Follow the steps in the rest of this document to take a measurement of ***5 minutes****, at* ***100 Hz.***



1.5.3 *Please send the data of each sensor to Sensequake to be checked prior to any site test.*

2. Synchronizing the Units

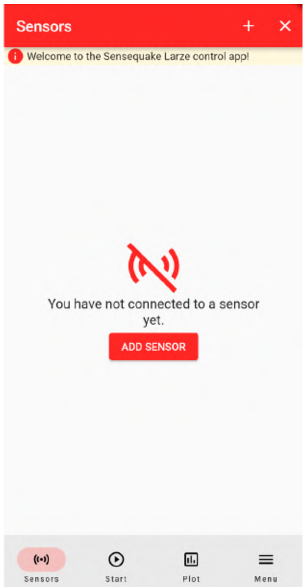
2.1 Take sensors out of their cases and attach the antennas to the ports on the side of each sensor unit.

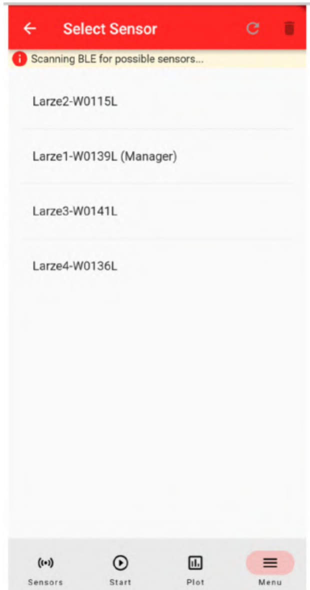
2.2 Power on each of the sensors starting with Sensor 1. Sensor LEDs should be flashing green if functioning properly.

2.3 To perform a “local sync” of the system, sensors must be within a close range similar to the following image. They should be within close range and have their antennas connected at this stage.

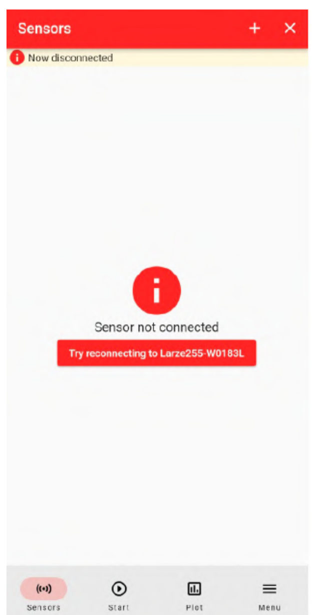
2.4 On the smartphone application, the “Manager” sensor should be selected to connect to. This sensor is numbered as Sensor 1. If this is the first time connecting, the **Sensors** tab will ask you to add a sensor. To do so:

- Menu > Connect to Sensor
- Select Larze1-XXXXXXX (Manager)



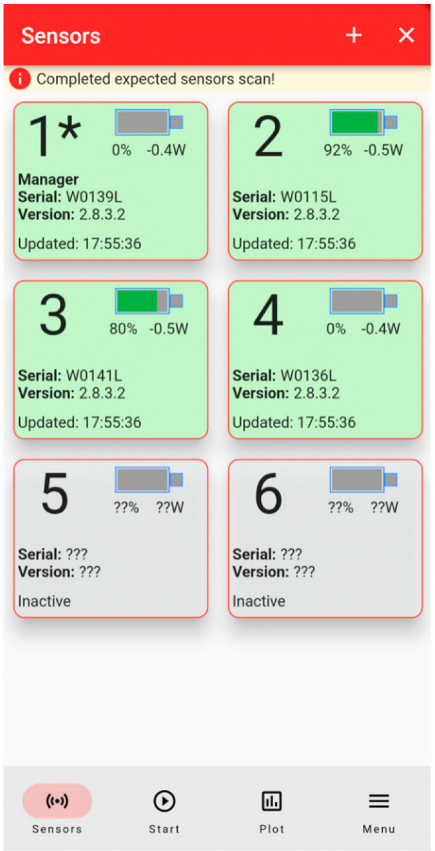


2.5 If the sensor has been previously connected, the following screen will appear under the **Sensors** tab. Click on Try reconnecting to Larze1-XXXXXXX to begin synchronization.



2.6 A list of all connected and synchronized sensors will appear. If sensors are missing from the list, press on the **+** icon at the top right of the page. You will be prompted to list the sensor numbers part of your network. The format should be as follows to show sensors 1 to 6, as an example: “2,3,4,5,6”.

The sensors are successfully synchronized if all expected units show up in this list and are surrounded by a green box.



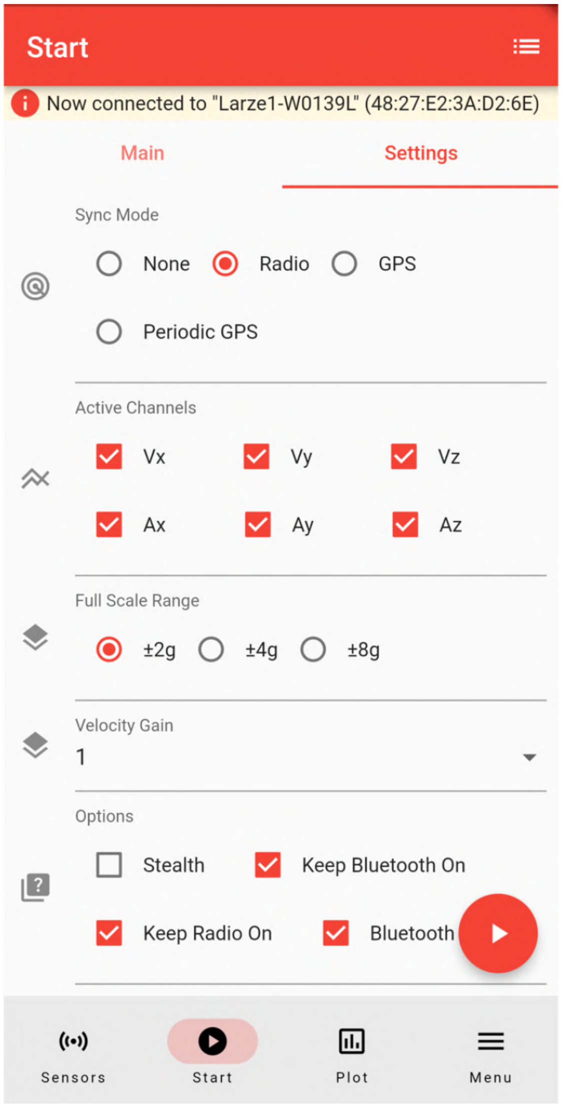
3. Synchronizing the Units

3.1 Once the sensors show up as green in the **Sensors** tab, navigate to the **Start** tab.

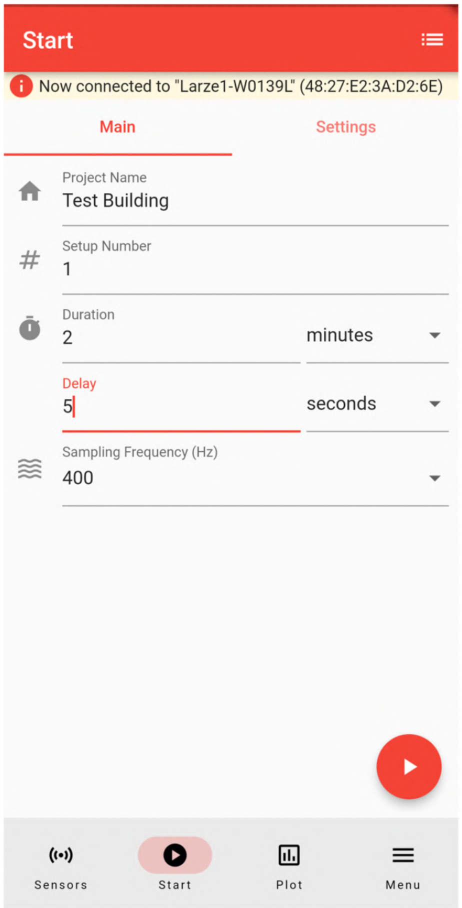
3.2 Before taking a measurement, the Settings sub-tab on this page should be reviewed. The following parameters need to be set to ensure proper data collection. These settings are saved unless changed in between setups:

- Sync mode should be set to Radio.
- All active channels should be checked off (Vx, Vy, Vz, Ax, Ay, Az)
- The full-scale range should be set to +/- 2g.

- The velocity gain should be set to 1.
- Under options, “Keep Bluetooth On”, “Keep Radio On” and “Bluetooth Realtime” should be selected.



3.3 Once the settings have been set, return to the Main Menu under the **Start** tab. Here, the test parameters shown on the Sensor layout should be input. The Delay parameter must be set depending on the time needed to bring the sensors from their Local Sync position to the proper position on the site. This may vary depending on the setup.



3.4 Once confirmed, the measurement command can be sent using the red Play Button icon at the bottom right of the screen.

4. Placing the sensors and data measurement

4.1 The sensors are ready to place at their designated locations once they each receive a solid blue LED. If the sensors have not all turned solid blue, they should be powered off and the process should be restarted.

The sensors will keep a solid blue light until the defined Delay is over. At this point, they will flash blue and collect data. The sensors should all be in the proper location at this stage. If not, the synchronization process should be repeated.

4.2 The sensors are ready to place at their designated locations once they each receive a solid blue LED. At this point, the antennas may be removed from the sensors if it facilitates transporting the units.

4.3 The following points should be diligently considered when placing the sensors at their correct pre-determined location:

- The sensors should be level with the surface. This should be done with the steel levelling plates and confirmed using a smartphone application or a bubble level if unsure. If the surface is level, the sensors can be placed directly on the surface.
- The surface should be free of any gravel or debris. There should be a solid and stable connection between the levelling plate and the surface of the bridge.
- The sensors' x-axes should all point in the same direction as specified on the sensor layout. This is normally parallel to the length of the bridge. The sensor should be aligned with the edge of the bridge as much as possible.
- If the sensor is close to an expansion joint, the sensor should be placed a few inches away from the joint. Below is an example of such a placement.

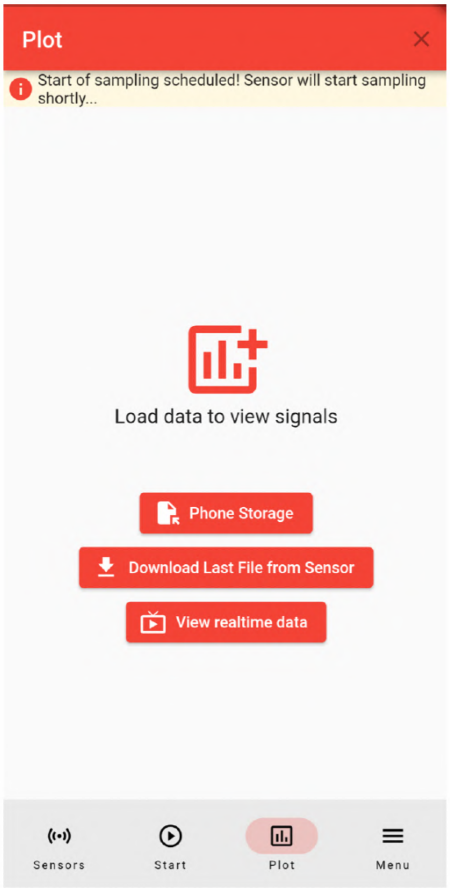
5. During the measurement

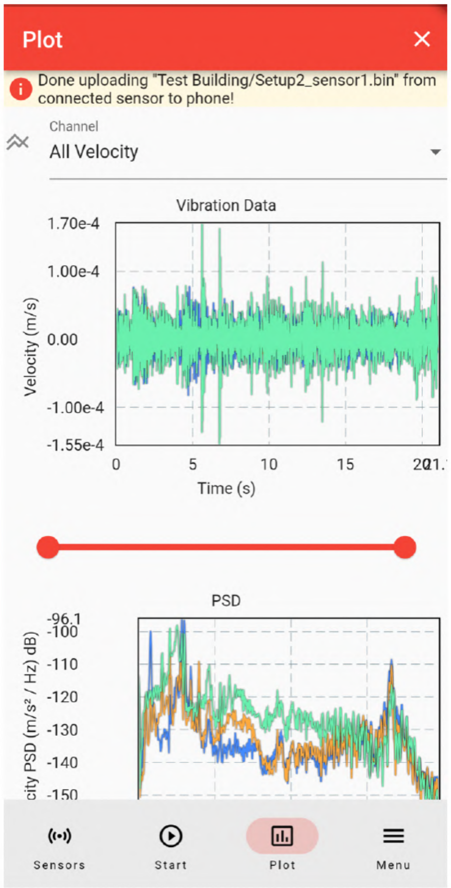
5.1 The sensors will begin collecting data once the blue LED begins to flash. If the light is flashing before the sensors are in place, the measurement should be repeated (return to Step 2.5).

5.2 Once the sensors begin to collect data, do not walk too close to the sensors as this may skew the results.

5.3 During the measurement, photos should be taken of the sensors and their context. These should be zoomed out enough to show the surrounding context of the sensors. A few examples are shown below. Photos should be transferred to Sensequake after the test named after the sensor numbers in the image and the setup number.

5.4 During the measurement, Sensor 1 can be connected in order to confirm that the measurements are being taken as well as their quality. To do this, approach Sensor 1 and go to the **Plot** tab. Once connected, select “All Velocity” to see the real-time data stream. Here, it is important to confirm that there are peaks in the “PSD” plot, especially below 20 Hz. For the reference sensor, these should be similar between the different setups.





5.5 Once the measurement is complete, the sensor lights will return to their flashing green state. At this point, the sensors can be brought back together for a Local Sync and the next setup can be undertaken (Step 2.5 and following).